Event Processing

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Understanding how GCXONE processes events is essential for operators and administrators. Every alarm you see on the dashboard has passed through a structured processing pipeline — from the moment a camera detects movement to the point an operator takes action.

1. Event Lifecycle

Every event in GCXONE follows a defined lifecycle:

Detection: A camera or sensor detects an activity and sends a signal to the GCXONE platform via the Proxy Layer.

Ingestion: The platform receives the raw event and logs it with a timestamp, device ID, site, and customer reference.

AI Analysis: The event is passed through the Nova 99x AI engine, which classifies it as a Real Alarm, False Alarm, or Technical Event.

Routing: Confirmed real alarms are routed to the operator queue via Talos for action.

Resolution: The operator reviews, processes, and closes the event. All actions are logged in the Audit Trail.

2. Event Types

GCXONE classifies every incoming event into one of three categories:

Real Alarm: A genuine security event that requires operator attention, such as motion detection, line crossing, or intrusion.

False Alarm: An event triggered by non-threatening activity such as animals, weather, or lighting changes. Filtered out by Nova 99x to reduce operator workload.

Technical Event: A system-generated event triggered by Health Check, such as a camera going offline, a black screen, obstruction detection, or low light conditions.

3. AI Processing — Nova 99x

Before any event reaches an operator, it passes through Nova 99x — GCXONE's AI filtering engine. Nova 99x analyzes each event using behavioral analysis and image recognition to determine whether it represents a real threat.

This process reduces false alarm volume by up to 99%, freeing operators to focus only on events that matter. Filtering can be applied across thermal and optical image types.

4. Event Routing

Once classified, events are routed based on their type and priority:

Real Alarms are pushed to the operator queue in Talos, where they can be assigned, reviewed, and actioned according to configured workflows.

False Alarms are filtered out and stored for reporting and analytics. They do not reach the operator queue.

Technical Events trigger automated notifications and can initiate workflows such as calling a technician, sending an SMS or email, or escalating to a supervisor.

5. Event States

Every event moves through the following states during its lifecycle:

New: Event received and pending review.

Filtered: Classified as a false alarm by Nova 99x and removed from the operator queue.

Assigned: Allocated to a specific operator for action.

In Progress: Operator is actively reviewing the event.

Processed: Event has been reviewed and closed.

6. Event Data & Audit

Every event is stored with full metadata including timestamp, device and sensor reference, site and customer, event type and classification, operator actions, and resolution notes. This data is available for reporting, SLA validation, and compliance audits through the Audit Log.